

7. Testing and Deployment

ComicCraft Project Documentation

Purpose: Explains the local deployment procedure and a practical testing checklist derived from the uploaded ComicCraft documentation.

7.1 Deployment Objective

The source document focuses on local deployment using FastAPI. The objective is to verify that AI text generation, image generation, layout construction, and PDF export work together before considering future cloud deployment.

7.2 Prepare the Environment

```
python -m venv env
env\Scripts\activate
pip install -r requirements.txt
```

The documented requirements include FastAPI, Uvicorn, Google Generative AI, Hugging Face Diffusers, FPDF, and related dependencies.

7.3 Configure Environment Variables

```
GEMINI_API_KEY=your_gemini_api_key_here
HF_API_KEY=your-huggingface-api-key-here
```

The source recommends environment variables or a .env file loaded with python-dotenv for sensitive credentials.

7.4 Start the Server

```
uvicorn app.main:app --reload
```

The documented local application address is <http://127.0.0.1:8000>. FastAPI interactive API documentation is available at /docs.

7.5 Testing Checklist

- Open the homepage and confirm the input form loads.
- Submit a valid story prompt and confirm form processing.
- Verify that a 5-panel outline is generated.
- Verify narration and dialogue generation.
- Verify that images are generated and stored under static/panels.
- Verify that the comic preview displays panel-by-panel.

- Verify that PDF export creates a file under static/exports.
- Verify the export success page.
- Use /docs to test API endpoints such as /generate-comic/json and /test-image.

7.6 Expected Test URLs

URL	Use
http://127.0.0.1:8000	ComicCraft homepage
http://127.0.0.1:8000/docs	FastAPI interactive API documentation

7.7 Deployment Verification

Successful local verification means the complete documented workflow operates: input → AI outline → story → images → layout → PDF export → user confirmation.